Jullien Thur

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Education

University of Utah **Fall 2025 - Spring 2029**

Bachelor of Science in Mechanical Engineering – Cumulative GPA: 4.00

Emphasis in Robotics & Controls, Minor in Computer Science

- Dean's List Recipient | Fall 2025 | GPA: 4.00
- Research Interests: Medical Robotics, Machine Learning, Control Systems

Skills

Technical Skills - SolidWorks, AutoCAD, Git/GitHub, Cura, Bambu Studio

Programming Languages - MATLAB, C++, JavaScript, HTML/CSS

Certifications - Certified SolidWorks Associate (Mechanical Design) – Associate Level

Interpersonal Skills – Leadership in Team-Based Environments, Conflict Resolution, Attention to Detail

Projects

Single-Axis Linear Actuator & Trajectory Control System **August 2025 - December 2025**

- Optimized C++ firmware by implementing non-blocking asynchronous timers, enabling parallel operation of the linear actuator and reloading mechanisms.
- Applied inverse kinematics in MATLAB to calculate precise servo angles for an automated cannon, evaluating the system's maximum and minimum effective ranges.
- Developed a state estimation protocol combining mechanical wall-calibration with an IR sensor on a high contrast rail to continuously track the robot's real-time position.
- Established serial communication between an Arduino microcontroller and MATLAB to relay coordinate data, ensuring accurate automated alignment and firing.

Submersible Water Pump **January 2026 - April 2026**

- Designed and developed a 3D-printable centrifugal water pump in a 3-person team, integrating a 3W motor under strict material constraints.
- Applied GD&T principles to the impeller and multi-part casing to eliminate pressure dead zones, minimize internal friction, and prevent axis whip during operation.
- Conducted rapid prototyping by evaluating 3 physical iterations within 2 weeks, scaling the impeller by 250% to maximize fluid displacement.
- Leveraged SolidWorks to optimize wet performance, achieving a consistent 2.5 GPM flow rate that surpassed the 1 GPM project goal.

Personal Portfolio Website **March 2026 - Present**

- Developed a responsive personal web application using JavaScript and HTML to showcase engineering projects, CAD designs, and software capabilities.
- Deployed the live site utilizing GitHub Pages for reliable hosting, managing source code, and continuous updates through Git version control.

Extracurricular Activities

Murray High Robotics **August 2022 - June 2025**

Team Leader/Designer

- Initiated and secured departmental funding to establish a new, permanent extracurricular robotics program.
- Utilized CAD software to model early-stage prototypes, analyzing designs for spatial efficiency and cost-effectiveness before physical assembly.

Work Experience

Murray City School District **July 2022 - June 2025**

Cybercorp IT Intern

- Diagnosed and repaired hardware issues on student Chromebooks, including replacing faulty logic board transistors and damaged displays.
- Utilized ServiceNow and Excel databases to manage and resolve IT helpdesk tickets for a large student and faculty population.
- Deployed and configured projection systems, ensuring seamless technological integration into classrooms to maximize effective instructional time.

References provided upon request